COMP 3920SEF/S492F
Machine Learning

Introduction

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- Characteristics of ML models
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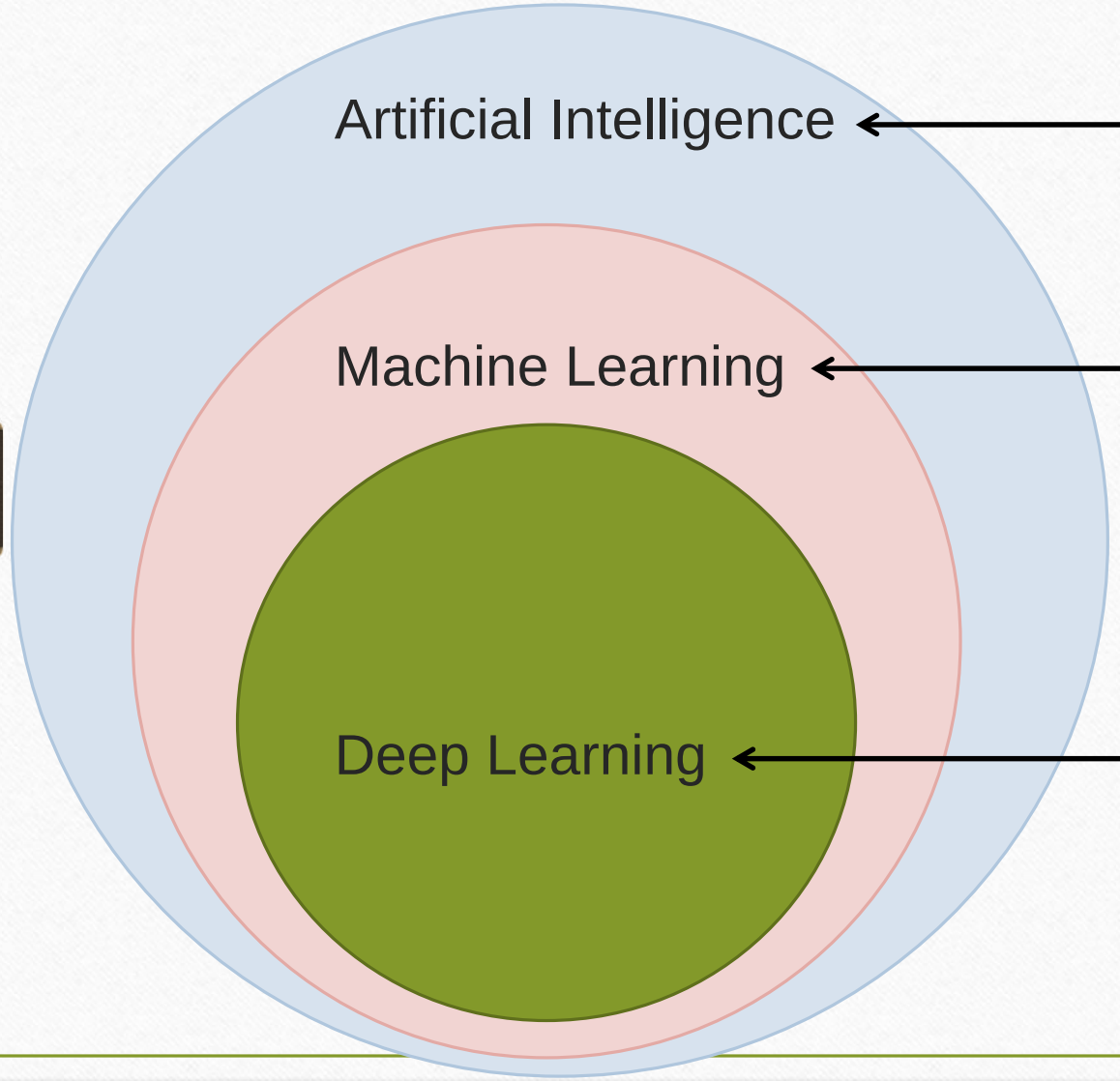
AI vs ML

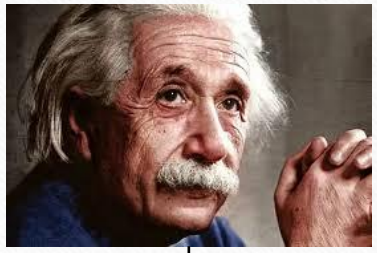


Any technique which enables machines to mimic human intelligence.

Machine learning uses statistical methods to enable machines to learn from data for predictions or decisions.

Deep learning uses many-layered neural networks to build algorithms that find the best way to perform tasks on their own, based on vast sets of data.

How machines learn?



rules and logic

Machine Evolved

input

output

input

output

find rules by itself
According to the given reward

Learning: Memorization vs true intelligence

examples

unseen examples

Learning: Memorization vs true intelligence

examples

Generalization

Learn one know three

unseen examples

Human learning vs machine learning

Human

- more sophisticated

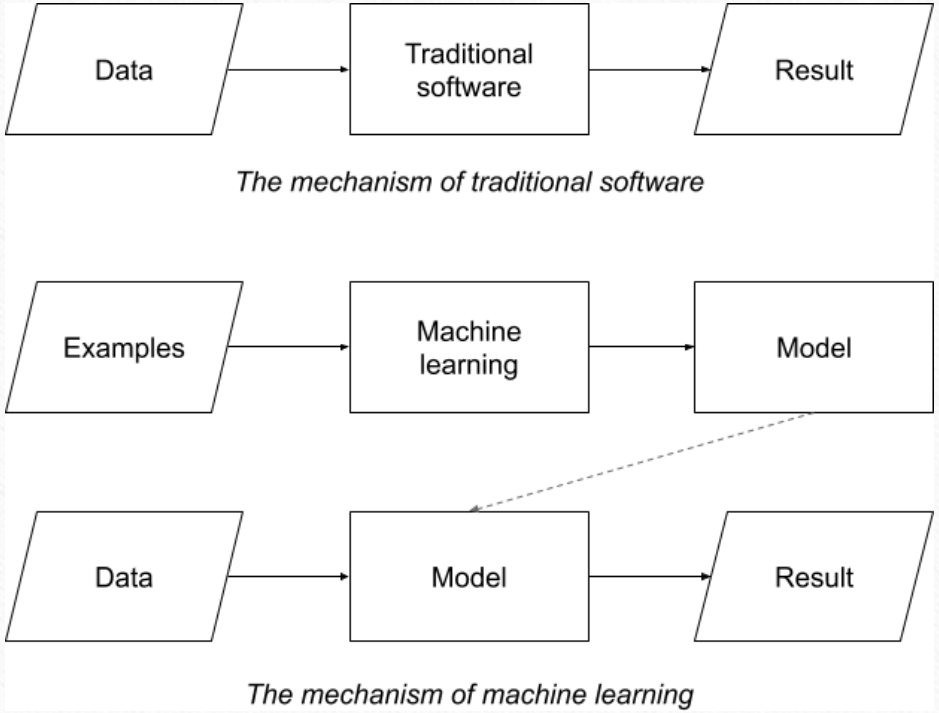
Machine

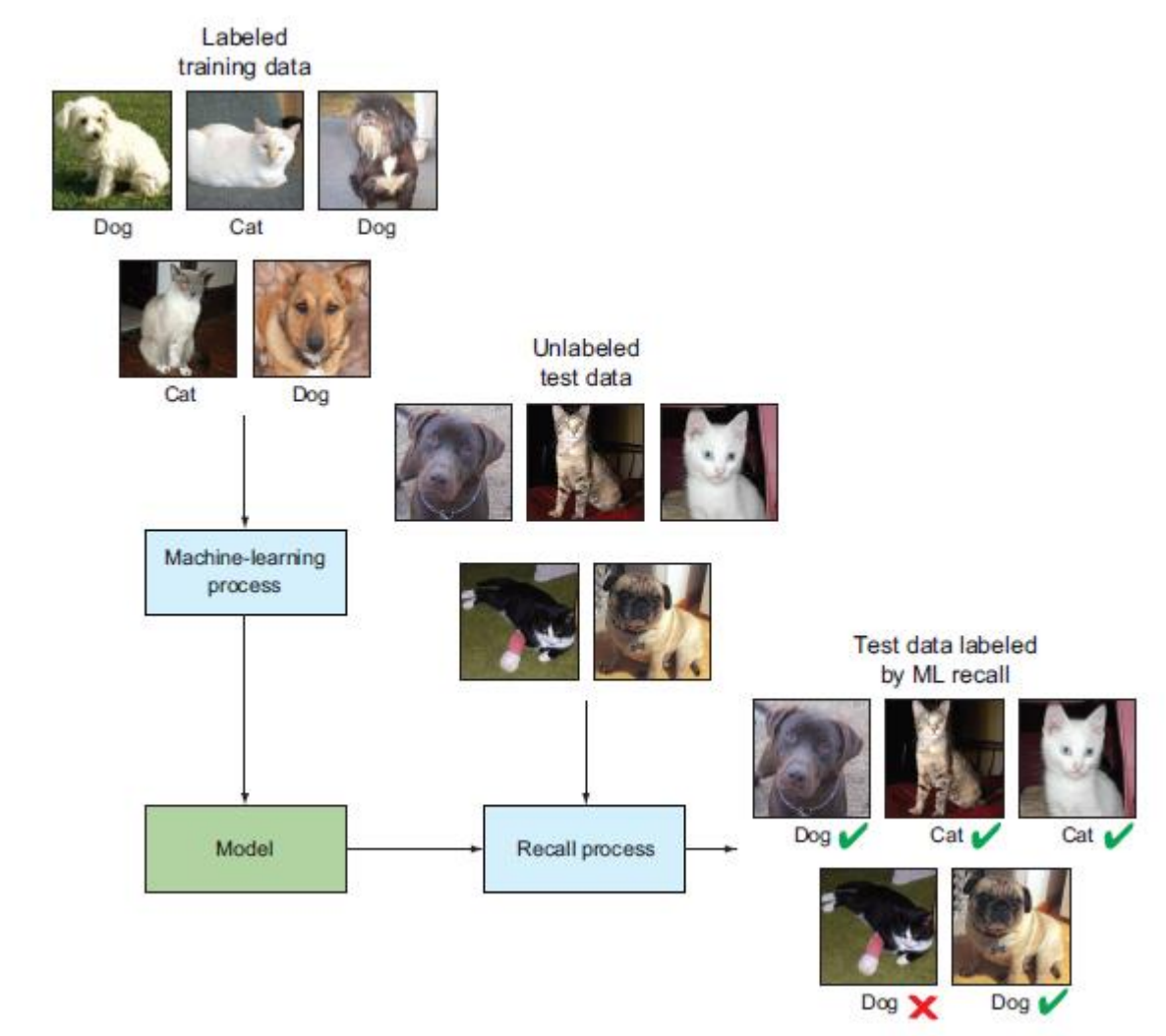
- greater capacity to memorize, recall, and process data

The ability to *generalize* and apply knowledge gained through training to new unseen examples is a key characteristic of both human and machine learning.
e.g. Google Quick Draw https://quickdraw.withgoogle.com/

Machine learning vs classical programming

- Learn from the past and predict the future
- Use data (1) to build a model which predicts useful results related to other data (2)
  - are examples or seen data
  - are unseen data

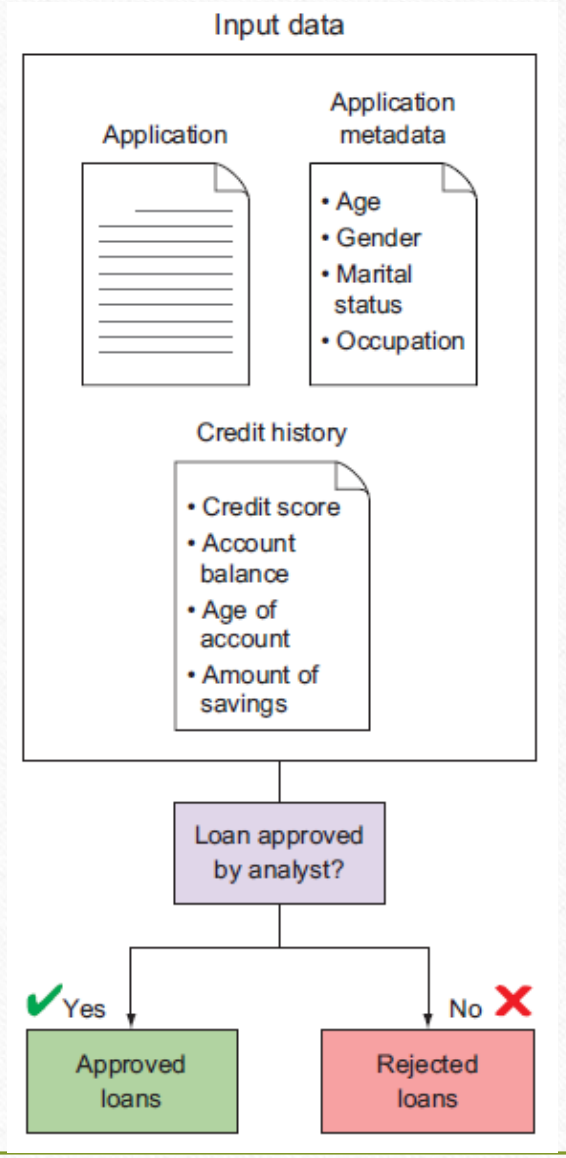


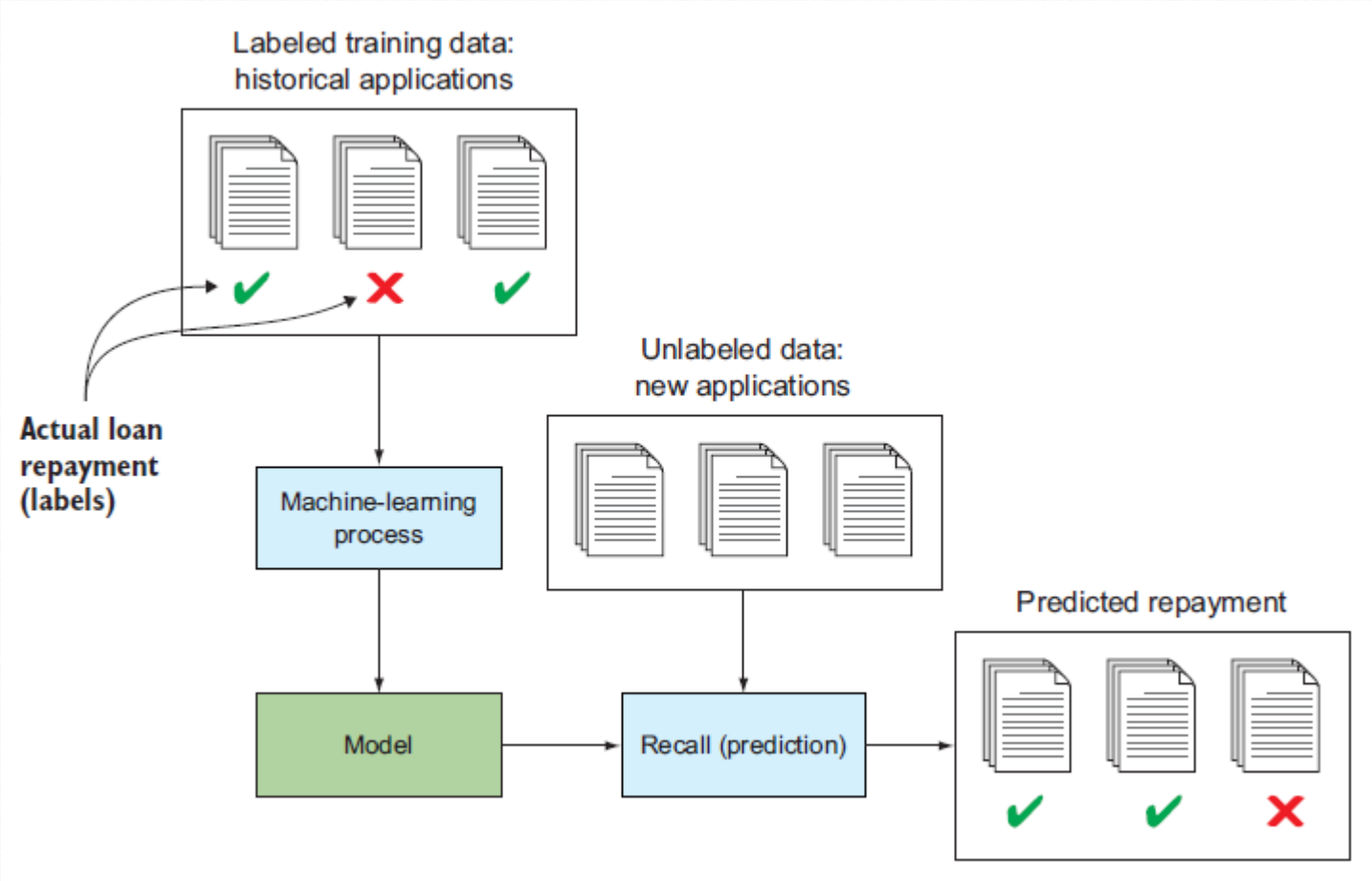


A machine-learning processes for the cats and dogs

Problem	Description	Example use cases
Classification	Determine the discrete class to which each individual belongs, based on input data	Spam filtering, sentiment analysis, fraud detection, customer ad targeting, churn prediction, support case flagging, content personalization, detection of manufacturing defects, customer segmentation, event discovery, genomics, drug efficacy
Regression	Predict the real-valued output for each individual, based on input data	Stock-market prediction, demand forecasting, price estimation, ad bid optimization, risk management, asset management, weather forecasting, sports prediction
Recommendation	Predict which alternatives a user would prefer	Product recommendation, job recruiting, Netflix Prize, online dating, content recommendation
Imputation	Infer the values of missing input data	Incomplete patient medical records, missing customer data, census data

Use cases for supervised machine learning, organized by the type of problem


Input data
Application
Application metadata
• Age
• Gender
• Marital status
• Occupation
Credit history
• Credit score
• Account balance
• Age of account
• Amount of savings
Loan approved by analyst?
Yes
No
Approved loans
Rejected loans


Labeled training data: historical applications
Actual loan repayment (labels)
Machine-learning process
Unlabeled data: new applications
Model
Recall (prediction)
Predicted repayment

Characteristics of ML models

Types of Machine Learning Models

1) Supervised
   - Classification (K-nearest, Logistic regression, ……)
   - Regression (Linear regression, ……)
   - ……
2) Unsupervised
   - Clustering (K-means, ……)
   - Dimensionality reduction (PCA, ……)
   - ……
3) Reinforcement
4) ……

It can be saved, loaded, and reused.

Model training

- *Model training*: determine the parameters and configurations to optimize the model of a certain type or structure.
- Use *training data.*
- May train multiple models to choose the best.

Model testing

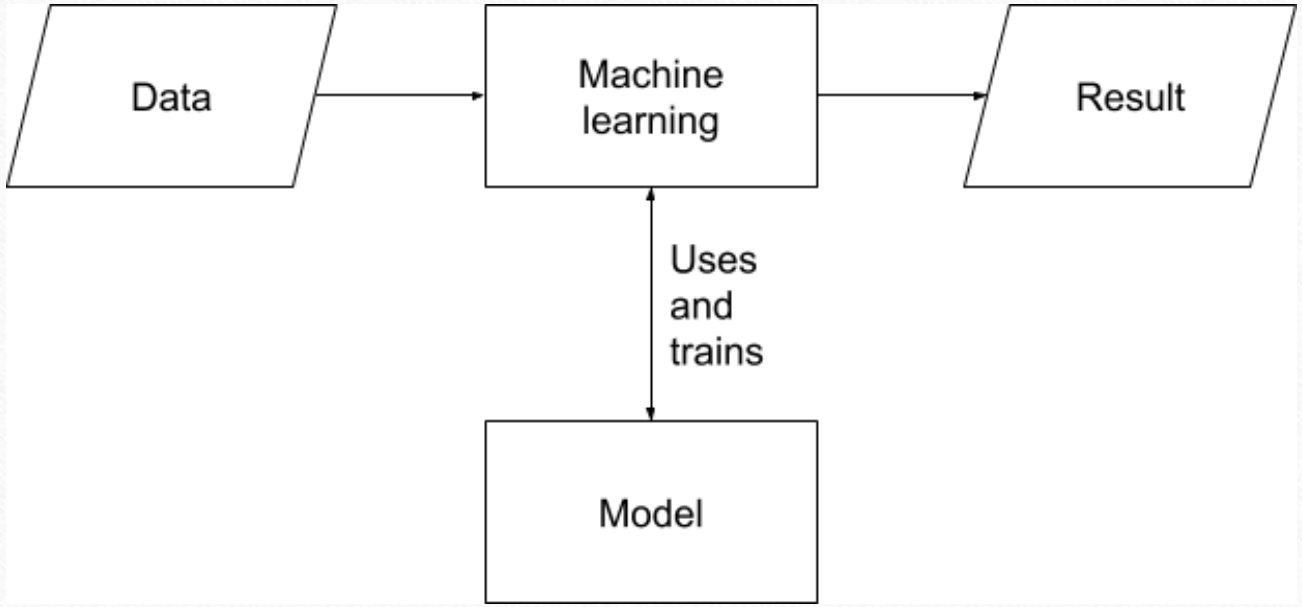
- *Model testing*: test the trained model to measure performance.
- Use *test data* (unseen)
- For checking if example data are enough, if the type or structure of the model is proper.

Model performance

- In general, more distinctive and representative examples improve model performance
- Model performance during training may differ from that during testing and using
  - A model may perform poorly during training, testing and using
  - A model may perform well during training, but poorly during testing and using

Continuous training

- In some cases, models may be updated or trained when being used (online)
- E.g. smartphone face ID, keyboard input



Pros and cons of using machine learning

Advantages

- **Accurate**—ML uses data to discover the optimal decision-making for your problem.
- **Fast**—ML can generate answers in a matter of milliseconds as new data streams in, allowing systems to react in real time.
- **Customizable**—Many data-driven problems can be addressed with machine learning.
- **Scalable**—As your business grows, ML easily scales to handle increased data rates.

Challenges

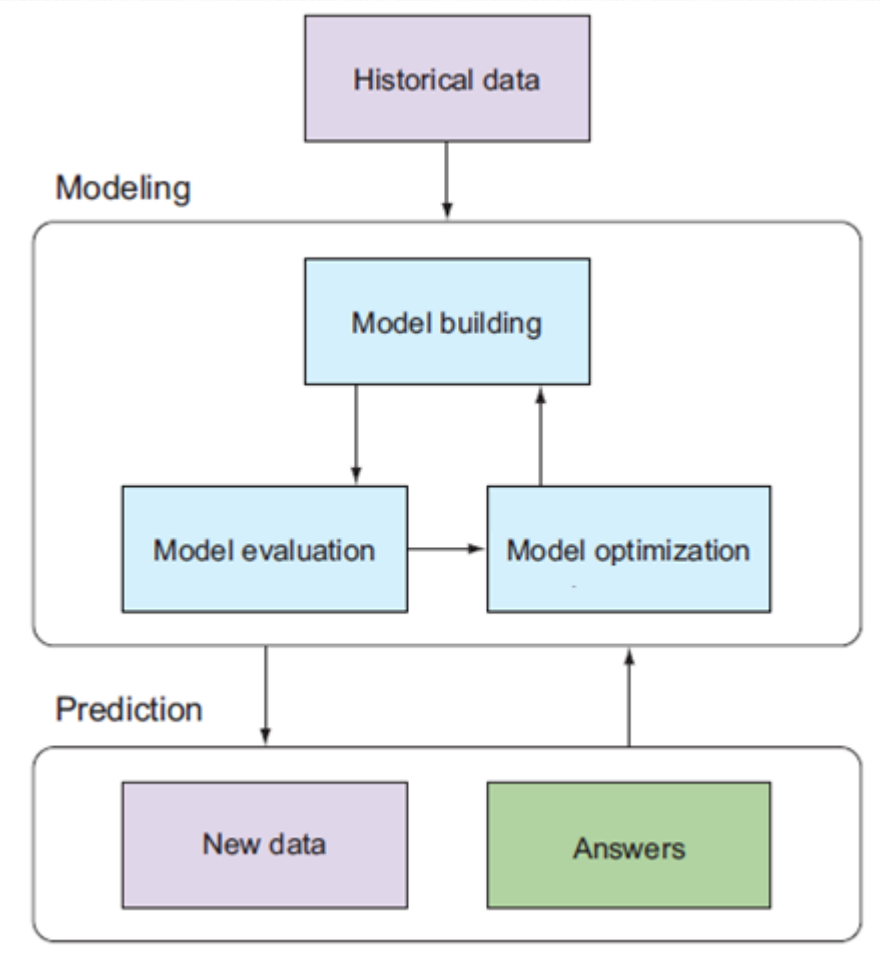
No.	*outlook*	*temperature*	*humidity*	*play*
x_1	sunny	L	H	*no*
x_2	?	H	H	*no*
x_3	overcast	M	?	*yes*
x_4	rainy	M	H	*yes*

- Data preparation
  - Cleaning: many real-world datasets contain missing values, outliers
  - Labeling: you need give label for each data sample
- Problem formulation
- Choosing a suitable algorithm
- Data Collection

ML workflow

Workflow

- Data collection and preparation
- Model building
- Evaluation
- Optimizing model performance
- Predictions on new data



Data collection and preparation

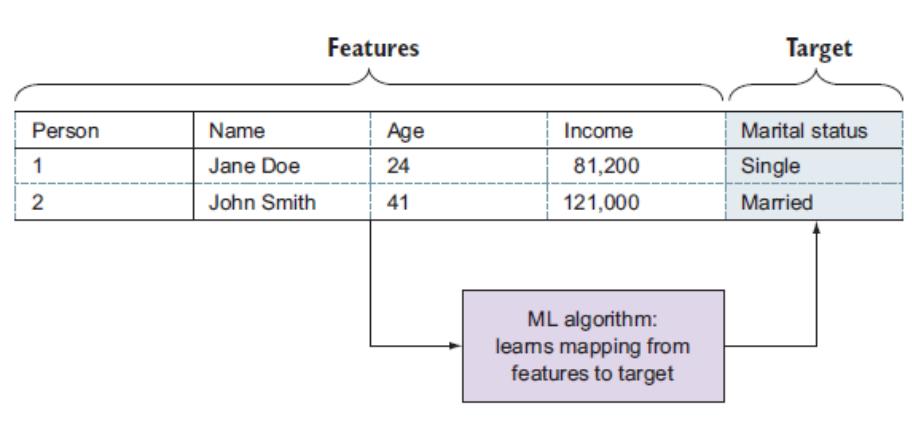
Features in columns

Person	Name	Age	Income	Marital status
1	Jane Doe	24	81,200	Single
2	John Smith	41	121,000	Married

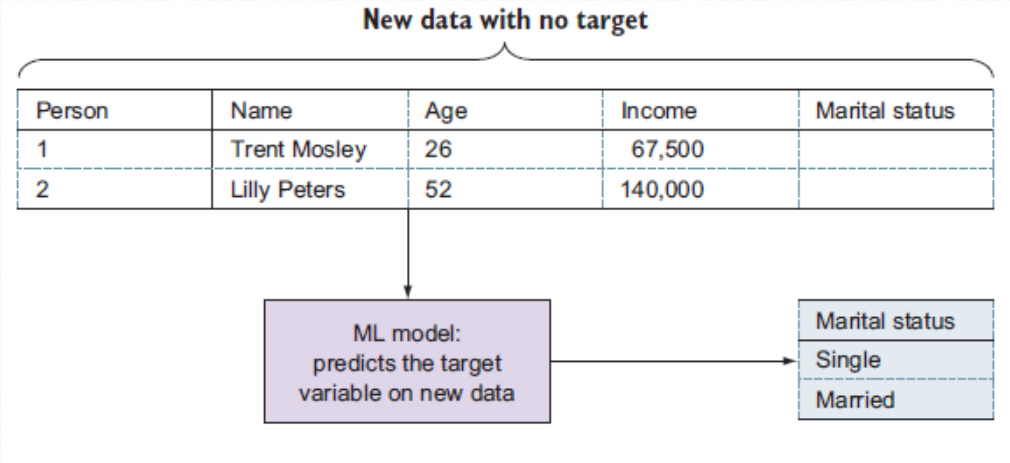
Examples in rows

In a tabular dataset, rows are called ***instances*** and columns represent ***features.***

Model building



The machine-learning modeling process

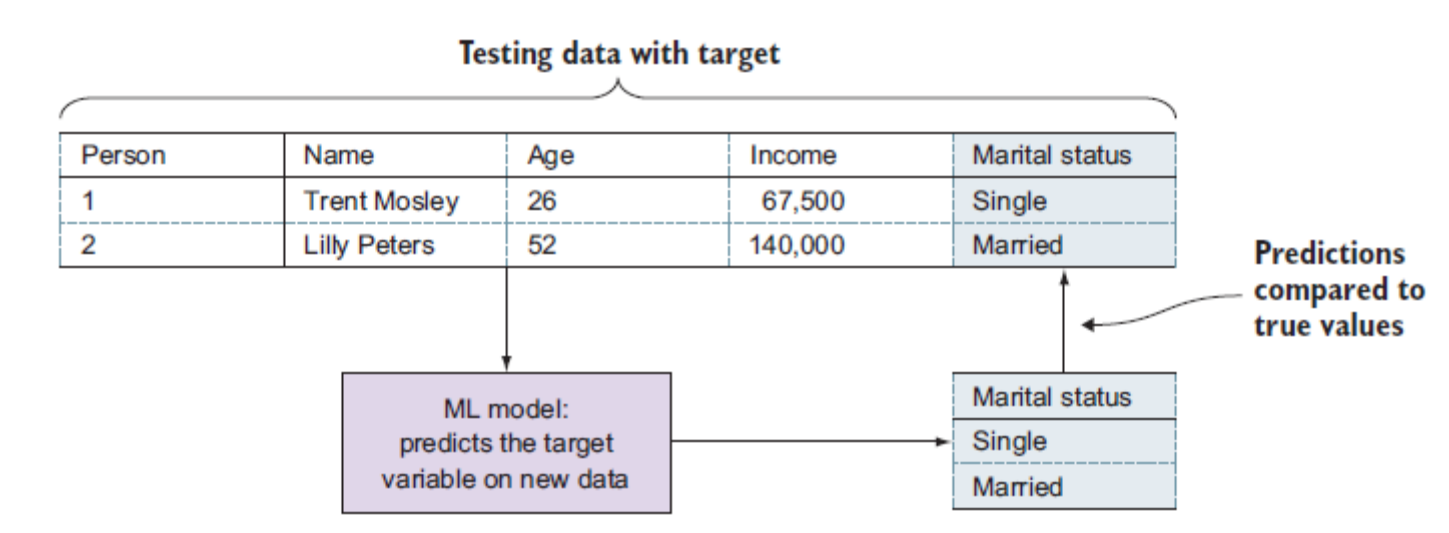


Using the model for prediction on new data

Listing 1.1 Initial structure of an ML workflow program

```
data = load_data("data/people.csv")
model = build_model(data, target="Marital status")
new_data = load_data("data/new_people.csv")
predictions = model.predict(new_data)
```

Evaluation



When using a testing set to evaluate model performance, you “pretend” that the target variable is unknown and compare the predictions with the true values.

Evaluation (cont.)

Listing 1.2 Our ML workflow program with model evaluation

```
data = load_data(...)
training_data, testing_data = split_data(data)
model = build_model(training_data, target="Marital status")
true_values = testing_data.extract_column("Marital status")
predictions = model.predict(testing_data)
accuracy = compare_predictions(predictions, true_values)
```

Optimizing model performance

- Tuning the model parameters
  - the optimal value of the parameters often depends on the type and structure of the data
- Selecting a subset of features and avoid noise
- Preprocessing the data

Common Terms

Word	Definition
instance or example	A single object, observation, transaction, or record.
target or label	The numerical or categorical (label) attribute of interest. This is the variable to be predicted for each new instance.
features	The input attributes that are used to predict the target. These also may be numerical or categorical.
model	A mathematical object describing the relationship between the features and the target.
training data	The set of instances with a known target to be used to fit an ML model.

Common Terms (cont.)

Word	Definition
supervised machine learning	Machine learning in which, given examples for which the output value is known, the training process infers a function that relates input values to the output.
unsupervised machine learning	Machine-learning techniques that don't rely on labeled examples, but rather try to find hidden structure in unlabeled data.
ML workflow	The stages in the ML process: data preparation, model building, evaluation, optimization, and prediction.

The end